VELMORA CO. · MANUFACTURING ERP

Production & Manufacturing Module

Complete Technical Documentation & Capstone Defense Specification

PROJECT TEAM	TARGET INDUSTRY	STACK	RELEASE DATE
Team 3 (Production)	Furniture & Woodcraft	Python 3.11, Flask, PostgreSQL	October 2026

1. Module Description

The **Production & Manufacturing Module** is the central workshop engine of the Velmora Co. Enterprise Resource Planning (ERP) platform. In a custom and batch furniture business, our biggest operational challenges are preventing workshop standstills, ensuring raw lumber and hardware are on hand before workers start cutting, and making sure that completed dining chairs or bedframes automatically show up in the inventory warehouse the moment they pass quality inspection.

Our module connects every step of the furniture manufacturing process: from cataloging furniture dimensions and setting up 1-unit Bill of Materials (BOM) recipes, to scheduling workshop batches, verifying material availability (MRP), dispatching inventory, tracking carpenter milestone progress with mobile photo proofs, and recording final QA approvals before rolling up direct job costs for Corporate Accounting.

> **Cross-Departmental Integrations in the Enterprise Database:**
>
> Our module operates with dedicated internal role-based access control (Production Manager vs. Workshop Floor Staff), while interfacing directly with two core partner systems in the enterprise database:
>
> - **Team 2 (Inventory & Warehousing):** Live bidirectional integration. We read live on-hand raw material quantities (`raw_material`) to calculate MRP shortage checks. When materials are dispatched to the workshop, our atomic database transactions deduct warehouse stock automatically. When finished furniture batches pass QA inspection, our module automatically credits finished goods stock (`finished_goods`) and catalog records (`item_master`) in real time. We also monitor zero-stock items to initiate restock batches.
> - **Team 5 (Finance & Cost Accounting):** Outbound job costing integration. We calculate the exact manufacturing cost for each completed batch (actual direct materials consumed, workshop labor hours logged, and standard 15% factory overhead) and publish the audited records to `prod_productioncost` for general ledger journal entries and inventory COGS valuation.

2. Objectives

2.1 General Objective

To build an automated, responsive, and traceable Production & Manufacturing ERP module for Velmora Co. that coordinates workshop floor activities, prevents raw material shortages, verifies assembly quality with photographic proof, and publishes accurate job costs to Corporate Accounting without manual double-entry.

2.2 Specific Objectives

1. **Standardize Recipe Formulation:** Allow managers to create Bill of Materials (BOM) recipes based on 1 finished unit, which dynamically scales across any batch size (e.g. 10 or 50 chairs).
2. **Prevent Floor Bottlenecks with MRP:** Implement Material Requirements Planning that compares batch demand against live warehouse inventory, alerting managers to shortages and locking order release until materials are verified.
3. **Safeguard Inventory with Atomic Transactions:** Guarantee that material stock deductions and finished goods additions happen inside single ACID database transactions, preventing negative or desynchronized stock balances.
4. **Enforce Physical Quality Assurance (QA):** Implement a 90% workshop gate requiring carpenters to upload client-compressed photos of their workmanship, which supervisors must review before approving stock intake.
5. **Automate Job Cost Accounting:** Compute direct material costs from dispatched inventory, record labor hours, apply a standard 15% factory overhead, and export finalized cost sheets to Accounting.
6. **Enforce Enterprise Separation of Duties (SoD):** Restrict order creation, recipe edits, material releases, and financial costing to the Production Manager, while allowing floor staff to view assigned batches and log build milestones.
7. **Deliver Low-Latency Cloud Access:** Use persistent connection pooling and in-memory caching to eliminate remote SSL handshake delays to our cloud database.

3. Functional Requirements (FR)

The functional requirements describe every feature and user action across the 11 sidebar tabs of the Velmora Production ERP.

Req ID	System Screen	Role	User Input / Trigger	System Logic & Database State Change
FR-AUTH-01	Login Screen	**BOTH**	Enters email/username and password.	Validates credentials against `public.users`. Verifies user belongs to Department 6 (`public.employee`). Resolves Position 6 as Manager and Position 2 as Staff.
FR-AUTH-02	Route Guard	**MANAGER**	Staff attempts to access Manager route.	Server decorator `@manager_required` blocks unauthorized requests, flashes an alert, and redirects to Dashboard.
FR-AUTH-03	Sign Out	**BOTH**	Clicks "Sign Out" in profile footer.	Destroys auth session keys while preserving flash messages. Sets `Cache-Control: no-store` to block back-button cache.
FR-DSH-01	Dashboard	**BOTH**	Dashboard load.	Aggregates active orders, completed batches, weekly activity bar chart, deadline tracking, and stage readiness bars.
FR-DSH-02	Dashboard	**MANAGER**	Stockout alert clicked.	Queries zero-stock items in `finished_goods` and links directly to order creation pre-filled with the stockout item.
FR-DSH-03	Dashboard	**BOTH**	Activity Timeline view.	Pulls the latest 4 live operational milestones (order creations, material dispatches, milestone advances, QC completions) using workshop terminology.
FR-PRD-01	Product Master	**MANAGER**	Add/Edit Furniture Modal.	Validates product dimensions and positive selling price; inserts or updates record in `prod_furniture`. (Staff has read-only view).
FR-BOM-01	Bill of Materials	**MANAGER**	Create Recipe Modal.	Initializes a 1-unit basis recipe header in `prod_bom` linked to a specific furniture blueprint.
FR-BOM-02	Bill of Materials	**MANAGER**	Add/Delete Material Item.	Inserts or removes line items in `prod_bomitem` with required quantities and units of measure. Recalculates unit standard cost.
FR-BOM-03	Bill of Materials	**STAFF**	Staff views recipe list.	System renders physical quantities and components but masks standard material unit costs and profit margins.
FR-ORD-01	Production Order	**MANAGER**	Create Order Modal.	Verifies BOM exists for product. Inserts batch order in `prod_productionorder` with status `Draft`.

Req ID	System Screen	Role	User Input / Trigger	System Logic & Database State Change
FR-ORD-02	Production Order	**MANAGER**	Warehouse Stockouts Restock Link.	Pre-populates selected zero-stock furniture model and automatically configures batch purpose as `Warehouse Restock`.
FR-PLN-01	Production Planning	**MANAGER**	Edit Plan Modal.	Sets planned start date, due date, and assigns a Department 6 craftsman. Dropdown displays active batch count per craftsman to prevent overloading.
FR-PLN-02	Production Planning	**BOTH**	Planning View load.	Evaluates batch health: `Needs Plan` (no dates), `On Schedule`, `Overdue`, or `Completed`.
FR-MRP-01	Material Requirement	**BOTH**	Requirement View load.	Multiplies BOM item quantity by planned batch quantity. Compares against live stock in `raw_material` and flags shortages.
FR-MRP-02	Material Requirement	**MANAGER**	Clicks "Release to Floor".	Validates 100% material availability. Updates order status from `Draft` to `Released`.
FR-ISS-01	Material Issuance	**MANAGER**	Clicks "Approve Dispatch".	Atomic transaction: subtracts raw materials from `raw_material`, logs dispatch in `prod_materialissuance`, seeds monitoring at 0%, sets status to `In Progress`.
FR-MON-01	Production Monitoring	**BOTH**	Update Progress Modal.	Updates progress milestone (25% Cutting, 50% Joinery, 75% Sanding, 90% Ready for QA) and logs operator notes in `prod_productionmonitoring`.
FR-MON-02	Production Monitoring	**BOTH**	Uploads stage photo.	Compresses photo client-side via HTML5 Canvas (max 1280px, 82% quality). Uploads to Supabase CDN bucket and saves URL in `prod_attachments`.
FR-QA-01	Finished Goods	**MANAGER**	Order reaches 90% build.	Order surfaces in QA queue. Manager inspects full-resolution photos in image lightbox viewer before approving.
FR-QA-02	Finished Goods	**MANAGER**	Selects "Passed".	Atomic transaction: logs pass in `prod_productionoutput`, sets progress to 100%, sets order to `Completed`, and upserts batch quantity into Team 2 `finished_goods`.
FR-QA-03	Finished Goods	**MANAGER**	Selects "Rework Required".	Logs rework reason, rolls back progress to 50% (Joinery & Assembly), increments rework counter, and sends batch back to workshop.
FR-CST-01	Production Costing	**MANAGER**	Record Cost Modal.	Rolls up direct material cost from issuances, calculates labor cost (hours × rate), adds 15% overhead, and commits cost sheet to `prod_productioncost` for Team 5.

Req ID	System Screen	Role	User Input / Trigger	System Logic & Database State Change
FR-REP-01	Production Reports	**MANAGER**	Reports Tab load.	Renders factory yield KPIs, order performance ledger, material variance analysis, and paginated transaction audit trail (Who, What, When).

4. Non-Functional Requirements (NFR)

4.1 Performance & Cloud Database Speed

- **Connection Socket Pooling:** Because our PostgreSQL database is hosted on a Supabase cloud instance in Australia, opening a new SSL connection on every web request takes 2.5 to 3.5 seconds. We implemented `psycopg2.pool.ThreadedConnectionPool`, cutting connection checkout time to under 1 millisecond.
- **In-Memory Catalog Caching:** Catalog tables that rarely change during a session (like furniture blueprints and raw material lists) are cached in memory in `cache.py` with automatic invalidation whenever items are updated.
- **N+1 Query Elimination:** In Material Requirements, component lookups are done against an in-memory dictionary rather than executing repetitive SQL queries inside loops.

4.2 Security & Internal Controls

- **Password Security:** User passwords are encrypted using bcrypt/PBKDF2 hashes. No plain text passwords exist in the database or server logs.
- **Server-Side Route Protection:** Modifying actions are protected with `@manager_required` decorators on the backend, ensuring that even if a staff member attempts to send direct POST requests to create orders or change recipes, the server halts the attempt.
- **Confidential Margin Masking:** Unit purchase costs, total BOM recipe costs, and profit markups are completely hidden from floor staff templates to protect company margins.
- **Session & History Protection:** When users log out, the server clears session keys and responds with `Cache-Control: no-store` to stop browsers from showing cached data when clicking the back button.

4.3 Data Integrity & Transaction Safety

- **ACID Multi-Table Transactions:** Complex workflows—like dispatching materials (which updates inventory, inserts issuance logs, initializes monitoring, and changes order status)—are wrapped inside a single atomic transaction. If any part fails, the entire transaction rolls back cleanly.
- **Zero Negative Stock Guarantee:** Inventory update queries enforce `AND stock_quantity >= %s`, raising an immediate error if stock is insufficient rather than allowing inventory to go negative.

4.4 Mobile Usability & Responsiveness

- **Responsive Layout (≤ 768px):** Built with mobile CSS media queries so supervisors and carpenters can use tablets or phones in the workshop. The top bar collapses into a clean hamburger menu and data tables support smooth touch horizontal scrolling.
- **Poka-Yoke Safeguards:** Crucial actions like order cancellations, QA rejections, and inventory issuances require deliberate modal confirmation to avoid accidental clicks.

5. Use Case Diagram

The use case diagram illustrates the roles and actions available to the **Production Manager** and **Production Staff**, as well as automatic integrations with partner department systems.

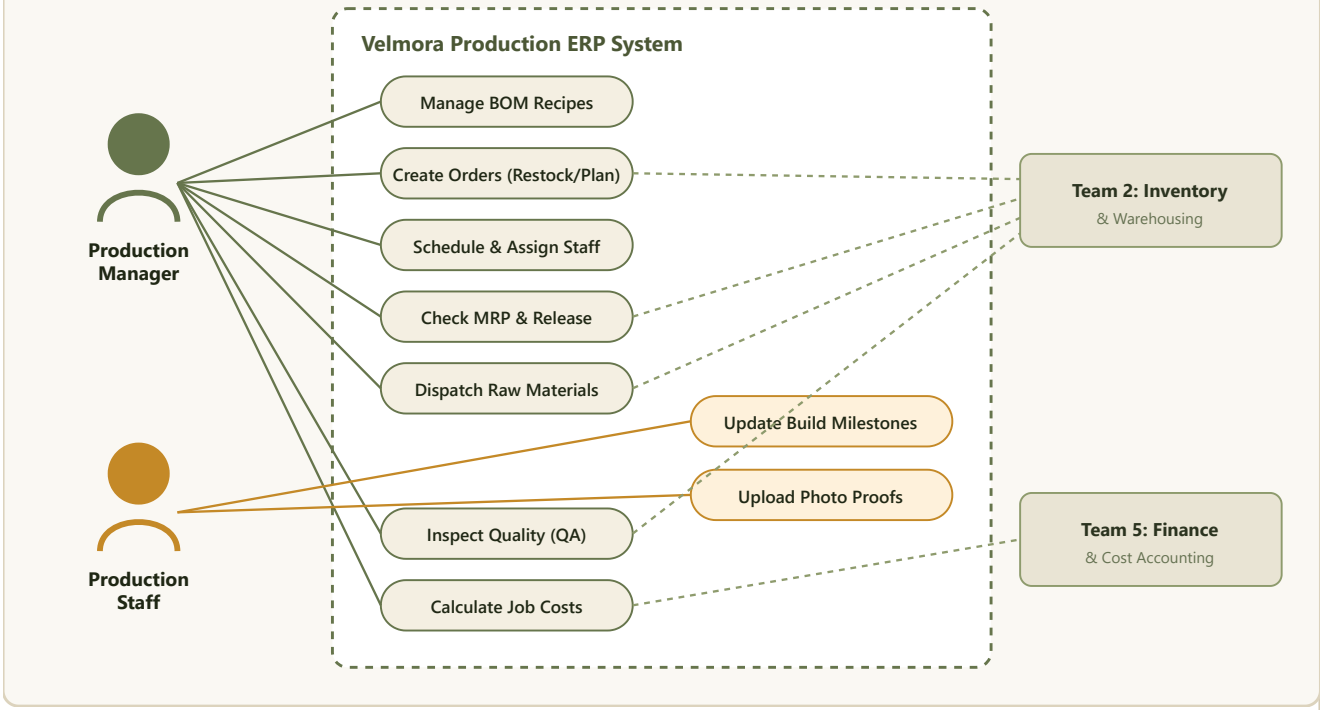


Figure 1: Use Case Diagram for Velmora Co. Production ERP Module

6. System Operational Flowchart

This flowchart illustrates the step-by-step lifecycle of an order from initialization, through feasibility testing and physical workshop milestones, up to final QA transfer and job costing.

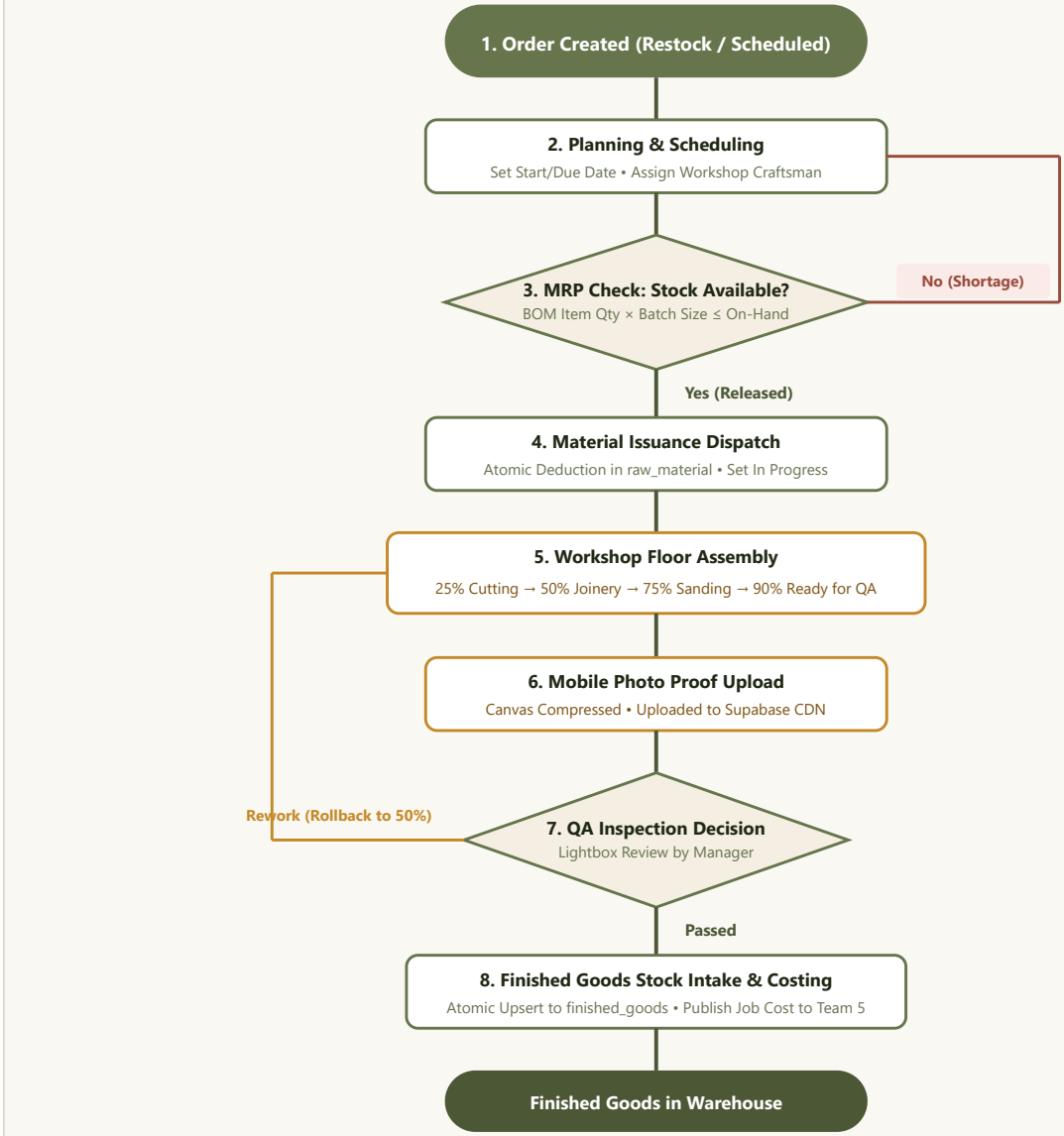


Figure 2: Operational Manufacturing Lifecycle Flowchart

7. Entity-Relationship Diagram (ERD)

The ERD shows how our internal `prod_` tables connect with shared enterprise tables (Inventory & Warehousing and Finance).

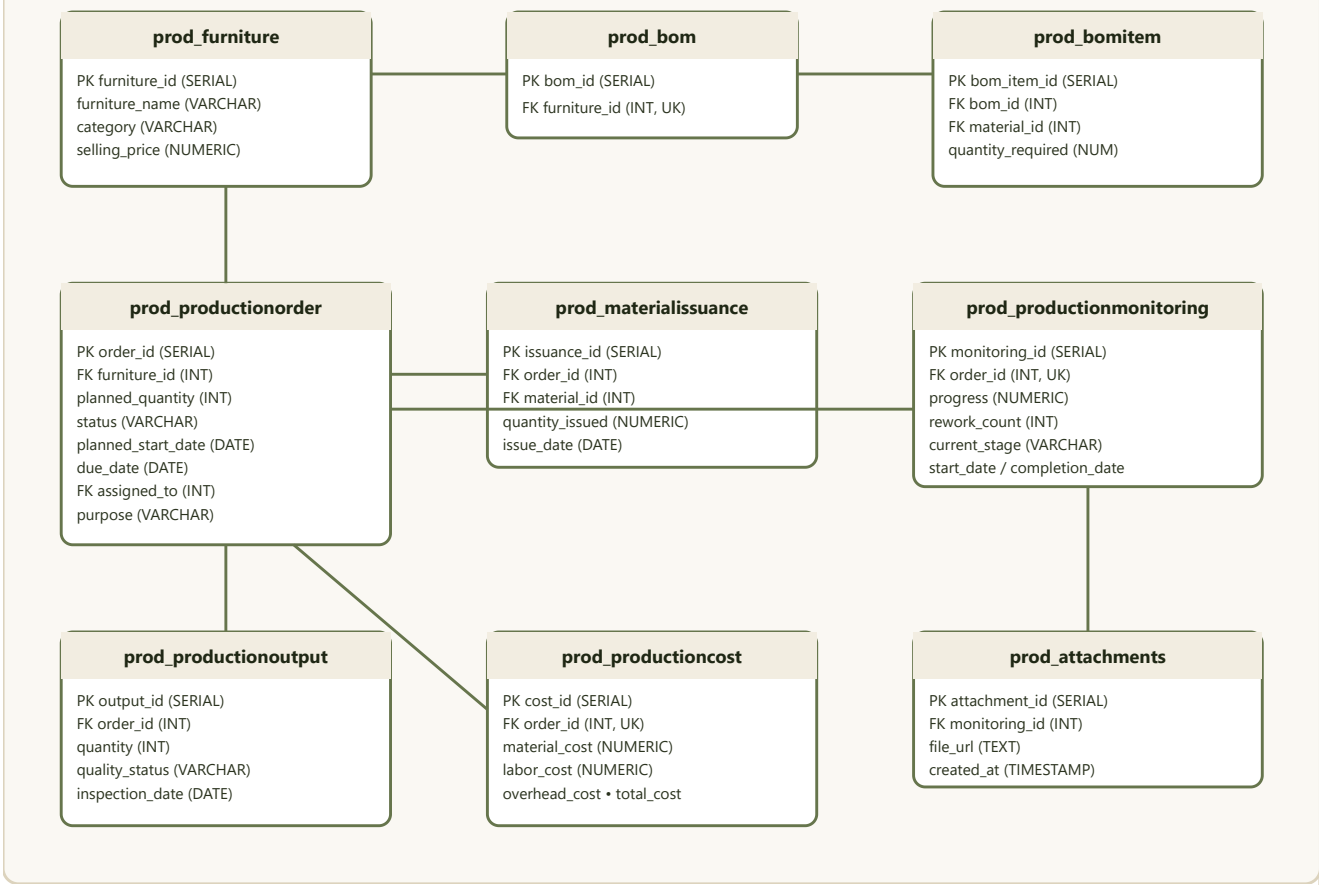


Figure 3: Relational Schema Architecture (ERD)

8. Comprehensive Data Dictionary

This data dictionary details the tables, columns, constraints, and real-world purposes of each database entity managed by the Production module.

8.1 `public.prod_furniture` (Product Blueprint Catalog)

Column Name	Type	Null?	Constraint	Description / Real-World Purpose
`furniture_id`	SERIAL	NO	PRIMARY KEY	Unique furniture blueprint identifier.
`furniture_name`	VARCHAR(150)	NO	NOT NULL	Commercial product name (e.g., Dining Chair, Oak Study Desk).
`category`	VARCHAR(50)	NO	NOT NULL	Product category: Seating, Tables, Storage, Beds, or Custom.
`dimensions`	VARCHAR(100)	YES	-	Standard workshop dimensions (Length × Width × Height in cm).
`selling_price`	NUMERIC(10,2)	NO	CHECK ≥ 0.00	Standard retail selling price recommended in catalog.
`image_url`	TEXT	YES	-	Public web link to product photography preview.

8.2 `public.prod_bom` (Recipe Header)

Column Name	Type	Null?	Constraint	Description / Real-World Purpose
`bom_id`	SERIAL	NO	PRIMARY KEY	Unique recipe header identifier.
`furniture_id`	INT	NO	UNIQUE, FK → prod_furniture	Enforces exactly 1 active master recipe per furniture model.

8.3 `public.prod_bomitem` (Recipe Ingredients)

Column Name	Type	Null?	Constraint	Description / Real-World Purpose
`bom_item_id`	SERIAL	NO	PRIMARY KEY	Unique line item ingredient identifier.
`bom_id`	INT	NO	FK → prod_bom (CASCADE)	Parent manufacturing recipe reference.
`material_id`	INT	NO	FK → raw_material	Raw material from Team 2 inventory (e.g. mahogany lumber, wood screws).
`quantity_required`	NUMERIC(10,2)	NO	CHECK > 0.00	Exact amount of material needed to build 1 finished unit.
`unit_of_measure`	VARCHAR(30)	YES	-	Unit of measure (e.g. board ft, pcs, liters, meters).

8.4 `public.prod_productionorder` (Batch Authorization)

Column Name	Type	Null?	Constraint	Description / Real-World Purpose
`order_id`	SERIAL	NO	PRIMARY KEY	Unique production order tracking number (#PO-1, #PO-2).
`furniture_id`	INT	NO	FK → prod_furniture	Furniture model authorized for production.

Column Name	Type	Null?	Constraint	Description / Real-World Purpose
`planned_quantity`	INT	NO	CHECK > 0	Number of units scheduled to be produced in this batch.
`status`	VARCHAR(30)	NO	DEFAULT 'Draft'	Lifecycle stage: Draft, Released, In Progress, Completed, Cancelled.
`order_date`	DATE	NO	DEFAULT CURRENT_DATE	Date when the order was registered.
`planned_start_date`	DATE	YES	-	Scheduled start date on workshop calendar.
`due_date`	DATE	YES	-	Target deadline for completion and workshop delivery.
`purpose`	VARCHAR(50)	NO	DEFAULT 'Warehouse Restock'	Manufacturing purpose: 'Warehouse Restock' (inventory replenishment) or 'Scheduled Batch' (planned production run).
`assigned_to`	INT	YES	FK → employee	Craftsman in Department 6 responsible for leading assembly.

8.5 `public.prod_materialissuance` (Inventory Dispatch Audit)

Column Name	Type	Null?	Constraint	Description / Real-World Purpose
`issuance_id`	SERIAL	NO	PRIMARY KEY	Unique dispatch voucher identifier (#IS-1, #IS-2).
`order_id`	INT	NO	FK → prod_productionorder	Order receiving the dispatched raw materials.
`material_id`	INT	NO	FK → raw_material	Raw material deducted from inventory stock.
`quantity_issued`	NUMERIC(10,2)	NO	CHECK > 0.00	Quantity of material delivered to the workshop floor.
`issue_date`	DATE	NO	DEFAULT CURRENT_DATE	Date when material transfer occurred.
`recorded_by`	VARCHAR(100)	NO	-	Name of the manager who approved warehouse dispatch.

8.6 `public.prod_productionmonitoring` (Shop-Floor Milestones)

Column Name	Type	Null?	Constraint	Description / Real-World Purpose
`monitoring_id`	SERIAL	NO	PRIMARY KEY	Unique workshop monitoring record identifier.
`order_id`	INT	NO	UNIQUE, FK → prod_productionorder	1-to-1 link to active production order.
`progress`	NUMERIC(5,2)	NO	CHECK 0.0 ≤ x ≤ 100.0	Current assembly milestone: 25%, 50%, 75%, 90%, 100%.
`rework_count`	INT	NO	DEFAULT 0	Number of times defects were sent back for workshop touch-up.
`current_stage`	VARCHAR(100)	YES	-	Workstation label (Cutting & Sizing, Joinery, Sanding, QA).
`operator_notes`	TEXT	YES	-	Floor remarks logged by carpenters.
`start_date`	DATE	YES	DEFAULT CURRENT_DATE	Date when physical work commenced.
`completion_date`	DATE	YES	-	Date when batch cleared 100% and passed QA inspection.

8.7 `public.prod_attachments` (Cloud Photo Proofs)

Column Name	Type	Null?	Constraint	Description / Real-World Purpose
`attachment_id`	SERIAL	NO	PRIMARY KEY	Unique photo proof identifier.
`monitoring_id`	INT	NO	FK → prod_productionmonitoring	Monitoring entry substantiated by the photo.
`file_url`	TEXT	NO	NOT NULL	Public CDN link to photo stored in Supabase Storage.
`file_name`	VARCHAR(255)	YES	-	Original device file name.
`created_at`	TIMESTAMP	NO	DEFAULT CURRENT_TIMESTAMP	Date and time when the photo was uploaded.

8.8 `public.prod_productionoutput` (QA Inspection Log)

Column Name	Type	Null?	Constraint	Description / Real-World Purpose
`output_id`	SERIAL	NO	PRIMARY KEY	Unique quality inspection report identifier (#QA-1).
`order_id`	INT	NO	FK → prod_productionorder	Order evaluated by Quality Inspector.
`quantity`	INT	NO	CHECK ≥ 0	Number of finished units inspected.
`quality_status`	VARCHAR(30)	NO	NOT NULL	Inspection decision: Passed, Rework Required, or Scrapped.
`inspection_date`	DATE	NO	DEFAULT CURRENT_DATE	Date when inspection took place.
`recorded_by`	VARCHAR(100)	NO	-	Name of the Quality Inspector.
`reason`	TEXT	YES	-	Remarks, defect description, or sign-off approval notes.

8.9 `public.prod_productioncost` (Job Manufacturing Cost Sheet)

Column Name	Type	Null?	Constraint	Description / Real-World Purpose
`cost_id`	SERIAL	NO	PRIMARY KEY	Unique job cost sheet reference (#CST-1).
`order_id`	INT	NO	UNIQUE, FK → prod_productionorder	1-to-1 link ensuring strict job costing integrity.
`material_cost`	NUMERIC(12,2)	NO	CHECK ≥ 0.00	Actual direct material expense calculated from issuances.
`labor_cost`	NUMERIC(12,2)	NO	CHECK ≥ 0.00	Direct workshop carpentry labor allocated to the batch.
`overhead_cost`	NUMERIC(12,2)	NO	CHECK ≥ 0.00	Standard 15% factory overhead (0.15 × (Material + Labor)).
`total_cost`	NUMERIC(12,2)	NO	CHECK ≥ 0.00	Total manufacturing investment (Material + Labor + Overhead).
`cost_date`	DATE	NO	DEFAULT CURRENT_DATE	Date when the cost sheet was calculated and finalized.